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## Practice questions

**Rubric:** Full credit may be given for any reasonable Bayesian analysis.

### 1) Probability of being sunny

From the following expressions, which one corresponds to the sentence: *The probability of being sunny given that it is June 14th of 1976?*

- (a)  $p(\text{sunny})$
- (b)  $p(\text{sunny} \mid \text{June})$
- (c)  $p(\text{sunny} \mid \text{June 14th of 1976})$
- (d)  $p(\text{June 14th of 1976} \mid \text{sunny})$
- (e)  $p(\text{sunny, June 14th of 1976})/p(\text{June 14th of 1976})$

**Answer:** The correct answers are **c** and **e** (the last one follows from the definition of conditional probability).

### 2) Picking a Pope

Show that the probability of choosing a human at random and picking the Pope is not the same as the probability of the Pope being human. In the animated series Futurama, the (Space) Pope is a reptile. How does this change your previous calculations?

**Answer:** Let's assume there are  $7.8 \cdot 10^8$  humans and there is only one Pope (Pope Leo XIV at the time of this writing). If a human is picked at random from the entire human population, the chances of that human being the pope are 1 in  $7.8 \cdot 10^8$ .

$$p(\text{Pope} \mid \text{human}) = \frac{1}{7.8 \cdot 10^8}$$

Because Pope are selected among humans. We can assert that given someone is the Pope, there is 100% chance he is a human. So,  $p(\text{human} \mid \text{Pope}) = 1$ .

Regarding the animated series Futurama and the space Pope is a reptile. We then have that  $p(\text{Pope} \mid \text{human}) = 0$  and  $p(\text{human} \mid \text{Pope}) = 0$ .



### 3) Broken telephone

Consider the following simplified version of the broken telephone game. A first player secretly communicates a message  $m_1 \in \{0, 1, 2\}$  to a second player with probabilities

$$P(m_1 = 0) = P(m_1 = 1) = P(m_1 = 2) = \frac{1}{3}.$$

The second player then secretly communicates a message  $m_2 = m_1 + n$  to a

third (final) player, where

$$P(n = 0) = \frac{2}{3}, \quad P(n = +1) = \frac{1}{3}.$$

(a) What is the optimal Bayes classification of the message received by the last player?

(b) What is the expected error probability?

**Answer:**

(a) Note that  $m_2 \in \{0, 1, 2, 3\}$  and

$$P(m_2 = m_1 \mid m_1) = \frac{2}{3} \quad P(m_2 = m_1 + 1 \mid m_1) = \frac{1}{3}$$

Hence the posterior distribution is:

$$P(m_1 = 0 \mid m_2) = \begin{cases} \frac{\frac{2}{3} \cdot \frac{1}{3}}{\frac{2}{3} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = 1, & m_2 = 0 \\ \frac{\frac{1}{3} \cdot \frac{1}{3}}{\frac{1}{3} \cdot \frac{1}{3} + \frac{2}{3} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = \frac{1}{3}, & m_2 = 1 \\ \frac{0 \cdot \frac{1}{3}}{0 \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3} + \frac{2}{3} \cdot \frac{1}{3}} = 0, & m_2 = 2 \\ \frac{0 \cdot \frac{1}{3}}{0 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3}} = 0, & m_2 = 3. \end{cases}$$

$$P(m_1 = 1 \mid m_2) = \begin{cases} \frac{0 \cdot \frac{1}{3}}{0 \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = 0, & m_2 = 0 \\ \frac{\frac{2}{3} \cdot \frac{1}{3}}{\frac{2}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = \frac{2}{3}, & m_2 = 1 \\ \frac{\frac{1}{3} \cdot \frac{1}{3}}{\frac{1}{3} \cdot \frac{1}{3} + \frac{2}{3} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = \frac{1}{3}, & m_2 = 2 \\ \frac{0 \cdot \frac{1}{3}}{0 \cdot \frac{1}{2} + 0 \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3}} = 0, & m_2 = 3. \end{cases}$$

and  $P(m_1 = 2 \mid m_2) = 1 - P(m_1 = 0 \mid m_2) - P(m_1 = 1 \mid m_2)$ .

Hence,

$$m_{1_{\text{pred}}}(m_2) = \begin{cases} 0, & m_2 = 0 \\ 1, & m_2 = 1 \\ 2, & m_2 = 2 \\ 2, & m_2 = 3. \end{cases}$$

$$(b) \ P(m_1 = 0 \mid m_2 = 1)P(m_2 = 1) + P(m_1 = 1 \mid m_2 = 2)P(m_2 = 2) = \frac{1}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3} = \frac{2}{9}$$

#### 4) Binary classification

Consider the following binary classification problem with losses  $L(C, C_{\text{pred}})$ :

|          | $C_{\text{pred}} = +1$ | $C_{\text{pred}} = -1$ |
|----------|------------------------|------------------------|
| $C = +1$ | 0                      | 100                    |
| $C = -1$ | 10                     | 0                      |

Your prior is

$$P(C = +1) = \frac{1}{10}.$$

- (a) What is the optimal classifier (and expected loss) based only on the prior distribution?
- (b) Suppose you can run an experiment with binary outcomes  $x \in \{0, 1\}$  with likelihoods

$$P(x = 1 \mid C = +1) = \frac{3}{4}, \quad P(x = 1 \mid C = -1) = \frac{1}{20}.$$

Find the classifier that minimizes expected loss. Compute the optimal expected loss and compare with part (a).

**Answer:**

- (a) The a priori Bayes classifier is  $C_{\text{pred}_{MAP}} = -1$  because  $P(C = +1) < P(C = -1)$ . However, this is not the classifier that minimizes expected loss since:

$$\begin{aligned}\mathbb{E}[L(C, C_{\text{pred}} = +1)] &= 10 P(C = -1) = 9 \\ &< \mathbb{E}[L(C, C_{\text{pred}} = -1)] = 100 P(C = +1) = 10\end{aligned}$$

The optimal expected loss is  $\mathbb{E}[L(C, C_{\text{pred}} = +1)] = 10P(C = -1) = 9$ .

- (b) The posterior distribution is:

$$P(C = +1 \mid x) = \begin{cases} \frac{\frac{3}{4} \cdot \frac{1}{10}}{\frac{3}{4} \cdot \frac{1}{10} + \frac{1}{20} \cdot \frac{9}{10}} = 0.625 & x = 1 \\ \frac{\frac{1}{4} \cdot \frac{1}{10}}{\frac{1}{4} \cdot \frac{1}{10} + \frac{19}{20} \cdot \frac{9}{10}} = 0.0284 & x = 0 \end{cases}$$

Hence,

$$\begin{aligned}\mathbb{E}[L(C, C_{\text{pred}} = +1) \mid x = 1] &= 10P(C = -1 \mid x = 1) = 3.75 \\ &< \mathbb{E}[L(C, C_{\text{pred}} = -1) \mid x = 1] = 100P(C = +1 \mid x = 1) = 62.5\end{aligned}$$

and

$$\begin{aligned}\mathbb{E}[L(C, C_{\text{pred}} = +1) \mid x = 0] &= 10P(C = -1 \mid x = 0) = 9.71 \\ &> \mathbb{E}[L(C, C_{\text{pred}} = -1) \mid x = 0] = 100P(C = +1 \mid x = 0) = 2.84\end{aligned}$$

Hence, the optimal classifier is:

$$C_{\text{pred}}(x) = \begin{cases} +1 & x = 1 \\ -1 & x = 0 \end{cases}$$

The expected loss under optimal classifier is:

$$\begin{aligned}\mathbb{E}[L(C, C_{\text{pred}}(x))] &= 10P(C = -1 \mid x = 1)P(x = 1) + \\ &\quad 100P(C = +1 \mid x = 0)P(x = 0) \\ &= 10(0.375)P(x = 1) + 100(0.0284)P(x = 0) \\ &= 10(0.375)(0.12) + 100(0.0284)(0.88) = 2.93 < 9\end{aligned}$$